NileCare - Network Design and Packet Tracer Configurations

Fictional lab for a GRC / network security practice project. 3 sites: Branch 1 (HQ), Branch 2, Branch 3. All names, IPs and passwords are for lab use only.

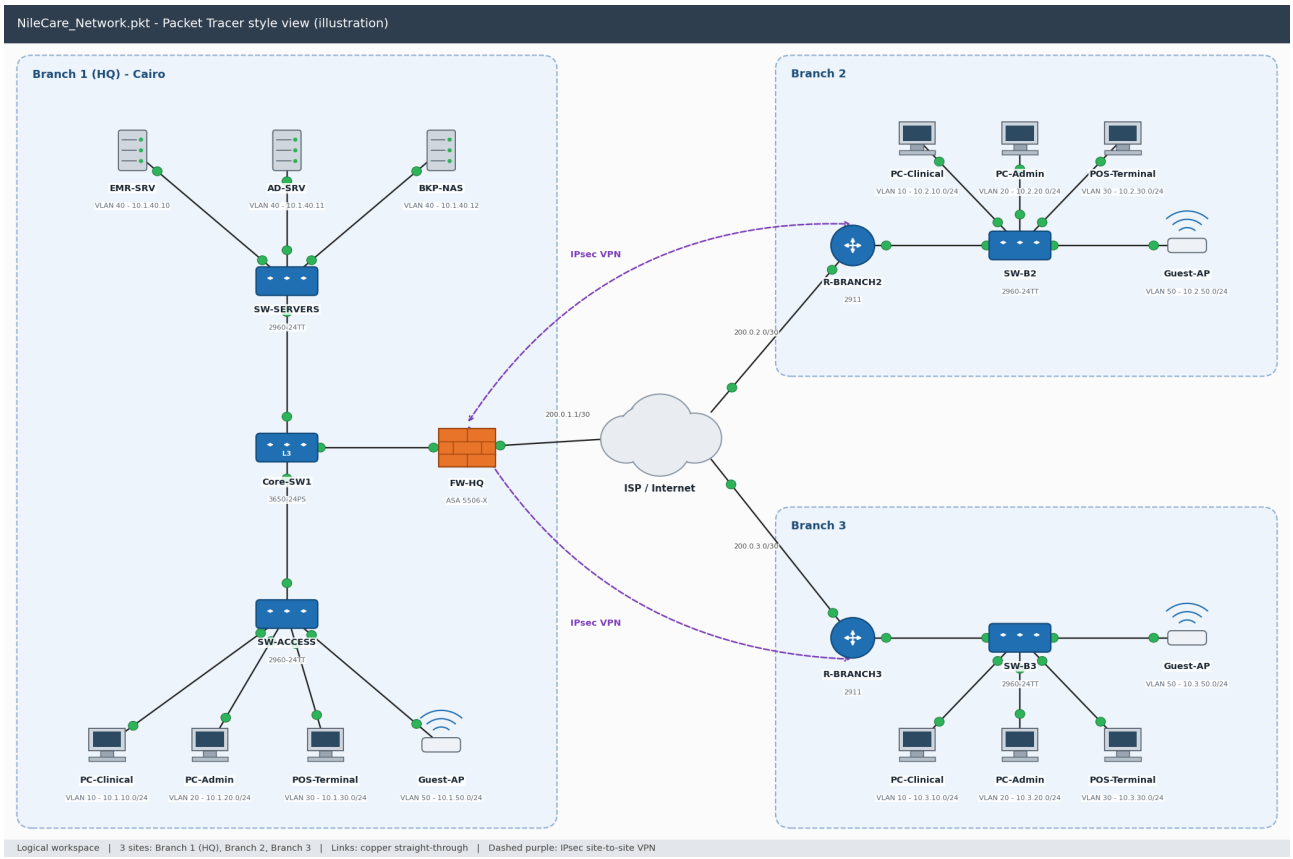


Illustration of the topology in Packet Tracer style. Build it in Packet Tracer with the devices and configs below and replace this picture with a real screenshot.

1. Devices (Packet Tracer models)

FW-HQ: ASA 5506-X | Core-SW1: 3650-24PS | SW-ACCESS, SW-SERVERS, SW-B2, SW-B3: 2960-24TT | ISP, R-BRANCH2, R-BRANCH3: 2911 | 3 servers, PCs, 3 Access Points. Use straight-through copper cables.

2. Addressing plan

VLAN	Name	Branch 1 (HQ)	Branch 2	Branch 3
10	Clinical	10.1.10.0/24	10.2.10.0/24	10.3.10.0/24
20	Admin	10.1.20.0/24	10.2.20.0/24	10.3.20.0/24
30	POS	10.1.30.0/24	10.2.30.0/24	10.3.30.0/24
40	Servers	10.1.40.0/24	-	-
50	Guest Wi-Fi	10.1.50.0/24	10.2.50.0/24	10.3.50.0/24
99	Management	10.1.99.0/24	-	-

WAN links: Core to FW 10.1.0.0/30 (.1 / .2) | FW to ISP 200.0.1.0/30 (.1 / .2) | ISP to Branch 2 200.0.2.0/30 (.1 / .2) | ISP to Branch 3 200.0.3.0/30 (.1 / .2) | ISP Loopback 8.8.8.8 (test internet).

Device	IP / gateway
EMR-SRV	10.1.40.10 /24, gw 10.1.40.1

Device	IP / gateway
AD-SRV	10.1.40.11 /24, gw 10.1.40.1
BKP-NAS	10.1.40.12 /24, gw 10.1.40.1
All PCs	DHCP (VLAN 10, 20, 30, 50)
Guest-AP	SSID NileCare-Guest, wired into VLAN 50 port

3. Cabling

From	Port	To	Port
Core-SW1	Gi1/0/1	SW-ACCESS	Gi0/1
Core-SW1	Gi1/0/2	SW-SERVERS	Gi0/1
Core-SW1	Gi1/0/24	FW-HQ	Gi1/1
FW-HQ	Gi1/2	ISP	Gi0/0
ISP	Gi0/1	R-BRANCH2	Gi0/0
ISP	Gi0/2	R-BRANCH3	Gi0/0
R-BRANCH2	Gi0/1	SW-B2	Gi0/1
R-BRANCH3	Gi0/1	SW-B3	Gi0/1
SW-ACCESS	Fa0/1 to Fa0/4	PC-Clinical, PC-Admin, POS, Guest-AP	Fa0
SW-SERVERS	Fa0/1 to Fa0/3	EMR-SRV, AD-SRV, BKP-NAS	Fa0
SW-B2 / SW-B3	Fa0/1 to Fa0/4	PC-Clinical, PC-Admin, POS, Guest-AP	Fa0

4. Baseline for every device (not the ASA)

Paste first, replacing the hostname. Lab password: NileCare@2026. Then apply the device config in the next sections.

```
enable
configure terminal
hostname <DEVICE-NAME>
no ip domain-lookup
service password-encryption
enable secret NileCare@2026
username admin secret NileCare@2026
ip domain-name nilecare.local
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
banner motd # Authorized access only. Activity is logged. #
line console 0
 logging synchronous
 login local
line vty 0 4
 login local
 transport input ssh
end
write memory
```

5. ISP router

```
hostname ISP
interface Loopback0
 description Test "internet" target
 ip address 8.8.8.8 255.255.255.255
interface GigabitEthernet0/0
 description To FW-HQ
 ip address 200.0.1.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 description To R-BRANCH2
 ip address 200.0.2.1 255.255.255.252
 no shutdown
interface GigabitEthernet0/2
 description To R-BRANCH3
 ip address 200.0.3.1 255.255.255.252
 no shutdown
end
write memory
```

6. Core-SW1 (3650)

If the switch rejects a command, check the port name with show ip interface brief. The 3650 does not need the trunk encapsulation command.

```
hostname Core-SW1
ip routing
vlan 10
 name CLINICAL
vlan 20
 name ADMIN
vlan 30
 name POS
vlan 40
 name SERVERS
vlan 50
 name GUEST
vlan 99
 name MGMT
! Trunks to access and server switches
interface range GigabitEthernet1/0/1 - 2
 switchport mode trunk
 no shutdown
! Routed port to the firewall
interface GigabitEthernet1/0/24
 description To FW-HQ
 no switchport
 ip address 10.1.0.1 255.255.255.252
 no shutdown
! Gateways (SVIs)
interface Vlan10
 ip address 10.1.10.1 255.255.255.0
interface Vlan20
 ip address 10.1.20.1 255.255.255.0
interface Vlan30
 ip address 10.1.30.1 255.255.255.0
 ip access-group POS_IN in
interface Vlan40
 ip address 10.1.40.1 255.255.255.0
interface Vlan50
 ip address 10.1.50.1 255.255.255.0
 ip access-group GUEST_IN in
interface Vlan99
 ip address 10.1.99.1 255.255.255.0
ip route 0.0.0.0 0.0.0.0 10.1.0.2
! Isolation: guest and POS cannot reach internal networks, internet allowed
ip access-list extended GUEST_IN
 deny ip 10.1.50.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
ip access-list extended POS_IN
 deny ip 10.1.30.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
! DHCP
ip dhcp excluded-address 10.1.10.1 10.1.10.9
```

```
ip dhcp excluded-address 10.1.20.1 10.1.20.9
ip dhcp excluded-address 10.1.30.1 10.1.30.9
ip dhcp excluded-address 10.1.50.1 10.1.50.9
ip dhcp pool CLINICAL
 network 10.1.10.0 255.255.255.0
 default-router 10.1.10.1
 dns-server 8.8.8.8
ip dhcp pool ADMIN
 network 10.1.20.0 255.255.255.0
 default-router 10.1.20.1
 dns-server 8.8.8.8
ip dhcp pool POS
 network 10.1.30.0 255.255.255.0
 default-router 10.1.30.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 10.1.50.0 255.255.255.0
 default-router 10.1.50.1
 dns-server 8.8.8.8
end
write memory
```

7. SW-ACCESS (2960)

```
hostname SW-ACCESS
vlan 10
 name CLINICAL
vlan 20
 name ADMIN
vlan 30
 name POS
vlan 50
 name GUEST
vlan 99
 name MGMT
interface GigabitEthernet0/1
 description Trunk to Core-SW1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,50,99
interface FastEthernet0/1
 description PC-Clinical
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
interface FastEthernet0/2
 description PC-Admin
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
interface FastEthernet0/3
 description POS-Terminal
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
interface FastEthernet0/4
 description Guest-AP
 switchport mode access
 switchport access vlan 50
interface range FastEthernet0/1 - 3
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
! Shut unused ports
interface range FastEthernet0/5 - 24
 shutdown
interface Vlan99
 ip address 10.1.99.11 255.255.255.0
 no shutdown
ip default-gateway 10.1.99.1
end
write memory
```

8. SW-SERVERS (2960)

```
hostname SW-SERVERS
vlan 40
 name SERVERS
vlan 99
 name MGMT
interface GigabitEthernet0/1
 description Trunk to Core-SW1
 switchport mode trunk
 switchport trunk allowed vlan 40,99
interface range FastEthernet0/1 - 3
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
interface range FastEthernet0/4 - 24
 shutdown
interface Vlan99
 ip address 10.1.99.12 255.255.255.0
 no shutdown
ip default-gateway 10.1.99.1
end
write memory
```

9. FW-HQ (ASA 5506-X) - base

```
hostname FW-HQ
interface GigabitEthernet1/1
 nameif inside
 security-level 100
 ip address 10.1.0.2 255.255.255.252
 no shutdown
interface GigabitEthernet1/2
 nameif outside
 security-level 0
 ip address 200.0.1.1 255.255.255.252
 no shutdown
route outside 0.0.0.0 0.0.0.0 200.0.1.2
route inside 10.1.0.0 255.255.0.0 10.1.0.1
! NAT: HQ networks to the internet
object network HQ-LAN
 subnet 10.1.0.0 255.255.0.0
 nat (inside,outside) dynamic interface
! Allow ping replies back in
policy-map global_policy
 class inspection_default
  inspect icmp
end
write memory
```

10. R-BRANCH2 (2911)

```
hostname R-BRANCH2
interface GigabitEthernet0/0
 description WAN to ISP
 ip address 200.0.2.2 255.255.255.252
 ip nat outside
 no shutdown
interface GigabitEthernet0/1
 description LAN trunk to SW-B2
 no ip address
 no shutdown
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 10.2.10.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 10.2.20.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/1.30
 encapsulation dot1Q 30
 ip address 10.2.30.1 255.255.255.0
 ip nat inside
 ip access-group POS_IN in
interface GigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 10.2.50.1 255.255.255.0
 ip nat inside
 ip access-group GUEST_IN in
ip route 0.0.0.0 0.0.0.0 200.0.2.1
ip access-list extended GUEST_IN
 deny ip 10.2.50.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
ip access-list extended POS_IN
 deny ip 10.2.30.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
! NAT, skipping VPN traffic to HQ
ip access-list extended NAT-ACL
 deny ip 10.2.0.0 0.0.255.255 10.1.0.0 0.0.255.255
 permit ip 10.2.0.0 0.0.255.255 any
ip nat inside source list NAT-ACL interface GigabitEthernet0/0 overload
ip dhcp excluded-address 10.2.10.1 10.2.10.9
ip dhcp excluded-address 10.2.20.1 10.2.20.9
ip dhcp excluded-address 10.2.30.1 10.2.30.9
ip dhcp excluded-address 10.2.50.1 10.2.50.9
ip dhcp pool CLINICAL
 network 10.2.10.0 255.255.255.0
 default-router 10.2.10.1
 dns-server 8.8.8.8
ip dhcp pool ADMIN
 network 10.2.20.0 255.255.255.0
 default-router 10.2.20.1
 dns-server 8.8.8.8
ip dhcp pool POS
 network 10.2.30.0 255.255.255.0
 default-router 10.2.30.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 10.2.50.0 255.255.255.0
 default-router 10.2.50.1
 dns-server 8.8.8.8
end
write memory
```

11. SW-B2 (2960)

```
hostname SW-B2
vlan 10
 name CLINICAL
vlan 20
 name ADMIN
vlan 30
 name POS
vlan 50
 name GUEST
interface GigabitEthernet0/1
 description Trunk to R-BRANCH2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,50
interface FastEthernet0/1
 description PC-Clinical
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
interface FastEthernet0/2
 description PC-Admin
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
interface FastEthernet0/3
 description POS-Terminal
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
interface FastEthernet0/4
 description Guest-AP
 switchport mode access
 switchport access vlan 50
interface range FastEthernet0/1 - 3
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
interface range FastEthernet0/5 - 24
 shutdown
end
write memory
```

12. R-BRANCH3 (2911)

```
hostname R-BRANCH3
interface GigabitEthernet0/0
 description WAN to ISP
 ip address 200.0.3.2 255.255.255.252
 ip nat outside
 no shutdown
interface GigabitEthernet0/1
 description LAN trunk to SW-B3
 no ip address
 no shutdown
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 10.3.10.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 10.3.20.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/1.30
 encapsulation dot1Q 30
 ip address 10.3.30.1 255.255.255.0
 ip nat inside
 ip access-group POS_IN in
interface GigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 10.3.50.1 255.255.255.0
 ip nat inside
 ip access-group GUEST_IN in
ip route 0.0.0.0 0.0.0.0 200.0.3.1
ip access-list extended GUEST_IN
 deny ip 10.3.50.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
ip access-list extended POS_IN
 deny ip 10.3.30.0 0.0.0.255 10.0.0.0 0.255.255.255
 permit ip any any
! NAT, skipping VPN traffic to HQ
ip access-list extended NAT-ACL
 deny ip 10.3.0.0 0.0.255.255 10.1.0.0 0.0.255.255
 permit ip 10.3.0.0 0.0.255.255 any
ip nat inside source list NAT-ACL interface GigabitEthernet0/0 overload
ip dhcp excluded-address 10.3.10.1 10.3.10.9
ip dhcp excluded-address 10.3.20.1 10.3.20.9
ip dhcp excluded-address 10.3.30.1 10.3.30.9
ip dhcp excluded-address 10.3.50.1 10.3.50.9
ip dhcp pool CLINICAL
 network 10.3.10.0 255.255.255.0
 default-router 10.3.10.1
 dns-server 8.8.8.8
ip dhcp pool ADMIN
 network 10.3.20.0 255.255.255.0
 default-router 10.3.20.1
 dns-server 8.8.8.8
ip dhcp pool POS
 network 10.3.30.0 255.255.255.0
 default-router 10.3.30.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 10.3.50.0 255.255.255.0
 default-router 10.3.50.1
 dns-server 8.8.8.8
end
write memory
```

13. SW-B3 (2960)

```
hostname SW-B3
vlan 10
 name CLINICAL
vlan 20
 name ADMIN
vlan 30
 name POS
vlan 50
 name GUEST
interface GigabitEthernet0/1
 description Trunk to R-BRANCH3
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,50
interface FastEthernet0/1
 description PC-Clinical
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
interface FastEthernet0/2
 description PC-Admin
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
interface FastEthernet0/3
 description POS-Terminal
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
interface FastEthernet0/4
 description Guest-AP
 switchport mode access
 switchport access vlan 50
interface range FastEthernet0/1 - 3
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
interface range FastEthernet0/5 - 24
 shutdown
end
write memory
```

14. IPsec VPN (do this last, after sections 1 to 13 work)

Hub and spoke: each branch has a tunnel to HQ. Branch-to-branch traffic is not part of this design. Packet Tracer's ASA may not accept the IPsec commands in some versions. If so, replace FW-HQ with a 2911 router using the same pattern as the branch VPN below (peer = the branch WAN address, crypto ACL 10.1.0.0/16 to 10.x.0.0/16).

FW-HQ:

```
object network B2-LAN
 subnet 10.2.0.0 255.255.0.0
object network B3-LAN
 subnet 10.3.0.0 255.255.0.0
! Do not NAT VPN traffic
nat (inside,outside) source static HQ-LAN HQ-LAN destination static B2-LAN B2-LAN
nat (inside,outside) source static HQ-LAN HQ-LAN destination static B3-LAN B3-LAN
access-list VPN-B2 extended permit ip 10.1.0.0 255.255.0.0 10.2.0.0 255.255.0.0
access-list VPN-B3 extended permit ip 10.1.0.0 255.255.0.0 10.3.0.0 255.255.0.0
crypto ikev1 policy 10
 authentication pre-share
 encryption aes
 hash sha
 group 2
 lifetime 86400
crypto ipsec ikev1 transform-set NC-TS esp-aes esp-sha-hmac
crypto map NC-MAP 10 match address VPN-B2
crypto map NC-MAP 10 set peer 200.0.2.2
crypto map NC-MAP 10 set ikev1 transform-set NC-TS
crypto map NC-MAP 20 match address VPN-B3
crypto map NC-MAP 20 set peer 200.0.3.2
crypto map NC-MAP 20 set ikev1 transform-set NC-TS
crypto map NC-MAP interface outside
```

```
crypto ikev1 enable outside
tunnel-group 200.0.2.2 type ipsec-l2l
tunnel-group 200.0.2.2 ipsec-attributes
 ikev1 pre-shared-key NileCare-VPN-Lab-Key
tunnel-group 200.0.3.2 type ipsec-l2l
tunnel-group 200.0.3.2 ipsec-attributes
 ikev1 pre-shared-key NileCare-VPN-Lab-Key
end
write memory
```

R-BRANCH2:

```
! Step 1 - enable the security license, then reload (save first)
license boot module c2900 technology-package securityk9
write memory
reload
! Step 2 - after reload
crypto isakmp policy 10
 encryption aes
 authentication pre-share
 group 2
 hash sha
 lifetime 86400
crypto isakmp key NileCare-VPN-Lab-Key address 200.0.1.1
crypto ipsec transform-set NC-TS esp-aes esp-sha-hmac
ip access-list extended VPN-TRAFFIC
 permit ip 10.2.0.0 0.0.255.255 10.1.0.0 0.0.255.255
crypto map NC-MAP 10 ipsec-isakmp
 set peer 200.0.1.1
 set transform-set NC-TS
 match address VPN-TRAFFIC
interface GigabitEthernet0/0
 crypto map NC-MAP
end
write memory
```

R-BRANCH3:

```
! Step 1 - enable the security license, then reload (save first)
license boot module c2900 technology-package securityk9
write memory
reload
! Step 2 - after reload
crypto isakmp policy 10
 encryption aes
 authentication pre-share
 group 2
 hash sha
 lifetime 86400
crypto isakmp key NileCare-VPN-Lab-Key address 200.0.1.1
crypto ipsec transform-set NC-TS esp-aes esp-sha-hmac
ip access-list extended VPN-TRAFFIC
 permit ip 10.3.0.0 0.0.255.255 10.1.0.0 0.0.255.255
crypto map NC-MAP 10 ipsec-isakmp
 set peer 200.0.1.1
 set transform-set NC-TS
 match address VPN-TRAFFIC
interface GigabitEthernet0/0
 crypto map NC-MAP
end
write memory
```

15. Tests

Test (from)	Command	Expected
PC-Clinical HQ	ping 10.1.40.10	Success
PC-Clinical HQ	ping 8.8.8.8	Success (NAT)
POS-Terminal HQ	ping 10.1.40.10	Fails (ACL POS_IN)
POS-Terminal HQ	ping 8.8.8.8	Success
PC-Clinical Branch 2	ping 10.1.40.10	Success (VPN)
PC-Clinical Branch 3	ping 10.1.40.10	Success (VPN)
POS-Terminal Branch 2	ping 10.2.10.1 or any 10.x host	Fails (ACL POS_IN)

Useful commands: show vlan brief | show ip interface brief | show ip route | show ip dhcp binding | show access-lists | show ip nat translations | show crypto isakmp sa | show crypto ipsec sa | ASA: show crypto ikev1 sa

16. Mapping to the gap assessment

Design element	ISO 27001:2022 Annex A
VLANs per function, guest and POS isolated by ACLs	A.8.22 Segregation of networks
Firewall, NAT, routed inside/outside zones	A.8.20 Networks security
IPsec VPN between sites	A.8.24 Use of cryptography; A.8.21 Network services
Local accounts, SSH only, no Telnet, password encryption	A.8.5 Secure authentication; A.8.9 Configuration management
Port security, unused ports shut down	A.7.2 Physical entry (network access); A.8.20
Management VLAN 99	A.8.9 Configuration management

Closing these gaps in the lab (segmentation, firewall, encryption, secure management) addresses risks R-01, R-09, R-10 and R-11 in the risk register.